

Operating System Project Proposal

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Backgrounds

- *Baleen: ML Admission & Prefetching for Flash Caches*
 - (FAST'24)
 - <https://www.usenix.org/conference/fast24/presentation/wong>
- Flash caches reduce load on HDD-backed storage, but limited flash endurance constrains the write rate.
 - Must choose valuable data

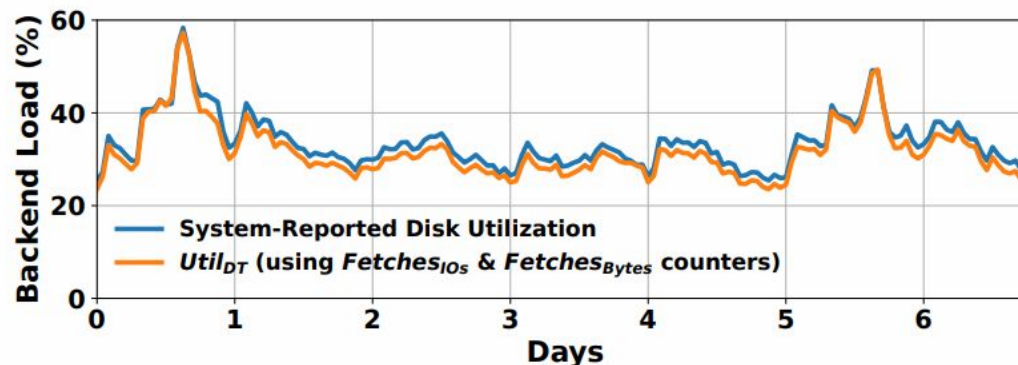
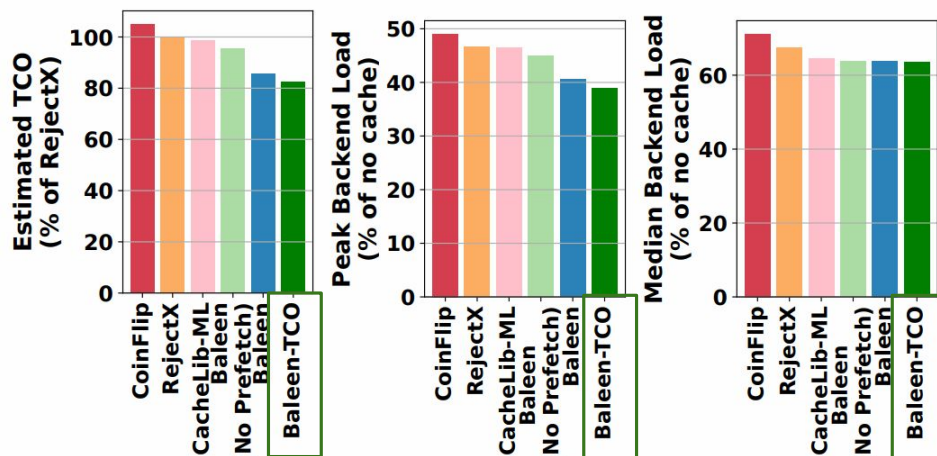
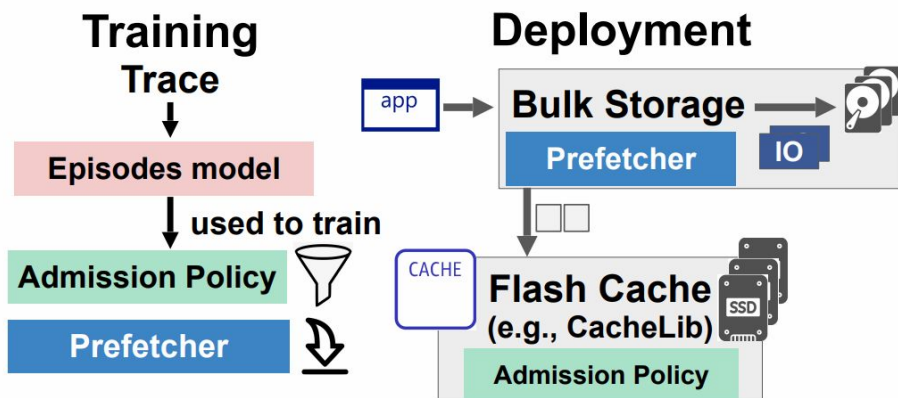


Figure 3: DT validated in production. Our DT formula (plugging counters into Eq 1) matches measured disk utilization (blue) closely. The peak of 58% occurs on Day 0.

Backgrounds

- *Baleen: ML Admission & Prefetching for Flash Caches*
- Baleen uses ML-based admission and prefetching to decide what to cache and what to prefetch.
 - They introduces episodes to model cache residency and train ML policies from an approximate optimal policy (OPT).
- Reduces peak HDD workload by 12% compared with prior cache policies.

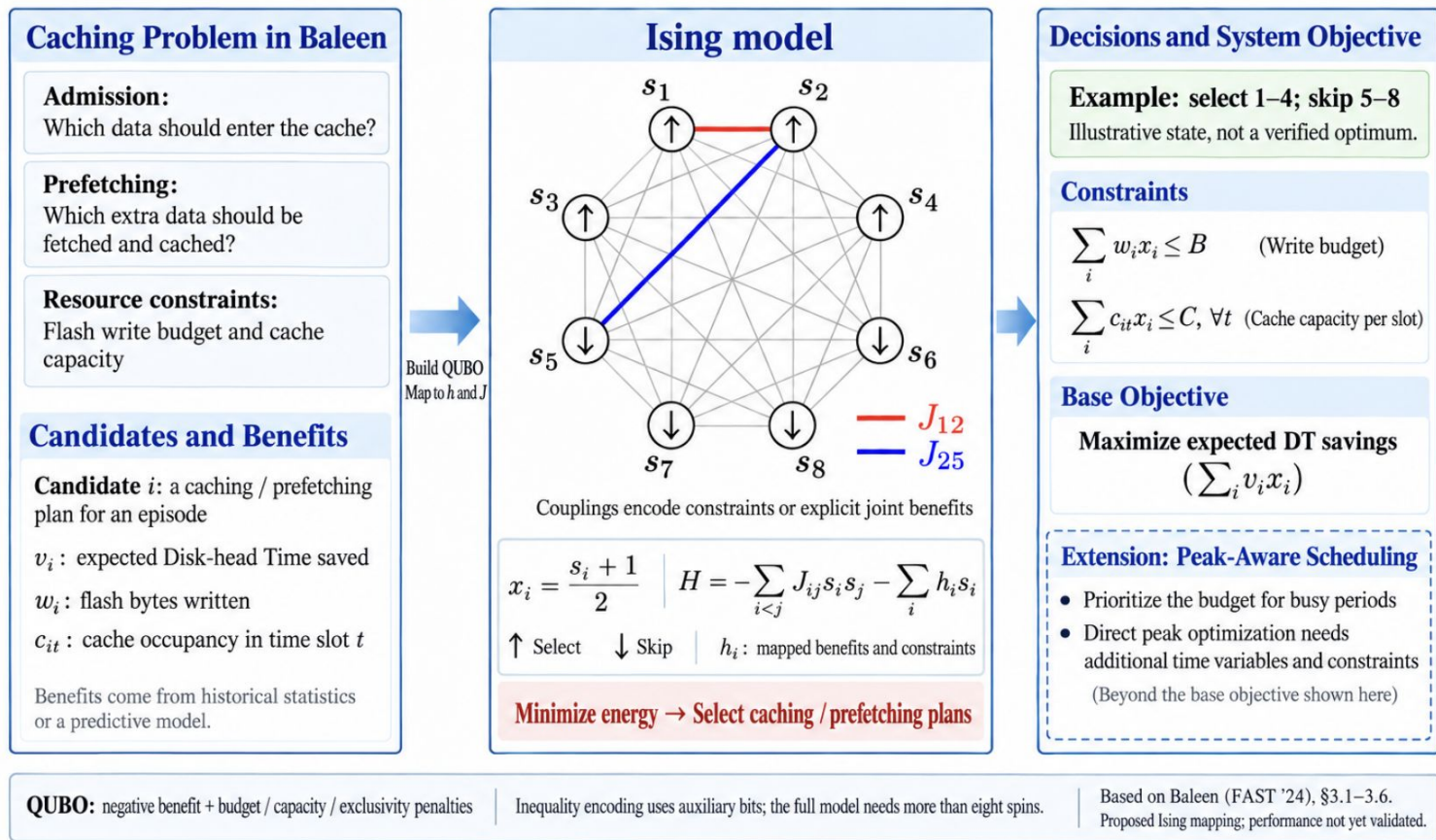


Gaps and Motivations

- Baleen Is Not Peak-Aware
- The system is evaluated on peak load, but its policy is not explicitly optimized for peak load.

Gaps and Motivations

- Baleen Is Not Peak-Aware
- Our question is: can we spend the same flash-write budget more efficiently by adopting Ising Machine?



Timeline

- Sep. 10 - 20: Literature review
- Sep. 21 - Oct. 5: Baseline & Problem Definition
- Oct. 6 - 20: Algorithm Design
- Oct. 21 - Nov. 1: Method Implementation & Analysis
- Nov. 2 - 15 — Full Evaluation & Finalize Results